2025

PR 10 MA1101 MATEMATIKA 1A

No.
Date:

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Jawaban

1 $y_0 = -x^2 + 3x$

$y_1 = 2x^3 - x^2 - 5x$

Pada $x \in [-2, 0]$, $y_0(x) \le y_1(x) \Rightarrow y_1 - y_0$

$x \in [0, 2]$, $y_0(x) \ge y_1(x) \Rightarrow y_0 - y_1$

Maka, luas dipecah menjadi 2

$$A = \int_{-2}^{0} y_1 - y_0 \, dx + \int_{0}^{2} y_0 - y_1 \, dx$$

$$= \int_{-2}^{0} 2x^3 - 8x \, dx + \int_{0}^{2} 8x - 2x^3 \, dx$$

$$= \frac{1}{2}x^4 - 4x^2 \Big|_{-2}^{0} + 4x^2 - \frac{1}{2}x^4 \Big|_{0}^{2}$$

$$= 8 + 8$$

$$= \boxed{16}$$

cara menuju x [karena y akar]

2 Dipecah menjadi dua bagian luas dengan

$L_1' = x - \frac{x^2}{4}$, $x \in [0, 1]$

$L_2' = 1 - \frac{x^2}{4}$, $x \in [1, 2]$

Gunakan integral

$$L_1 = \int_0^1 L_1' \, dx = \int_0^1 x - \frac{x^2}{4} \, dx$$

$$= \frac{1}{2}x^2 - \frac{1}{12}x^3 \Big|_0^1$$

$$= \frac{5}{12}$$

$$L_2 = \int_1^2 L_2' \, dx = \int_1^2 1 - \frac{x^2}{4} \, dx$$

$$= x - \frac{1}{12}x^3 \Big|_1^2$$

$$= \left(2 - \frac{8}{12}\right) - \left(1 - \frac{1}{12}\right)$$

$$= \frac{5}{12}$$

$$L_1 + L_2 = \boxed{\frac{5}{6}}$$

BisBOSS

Cara II tinjau dari sumbu y

atas $\Rightarrow y = \frac{x^2}{4}$

$\Rightarrow x = 2\sqrt{y}$

bawah $\Rightarrow y = x$

$\Rightarrow x = y$

y dari 0 sampai 1, maka

$\int_0^1 2\sqrt{y} - y \, dy$

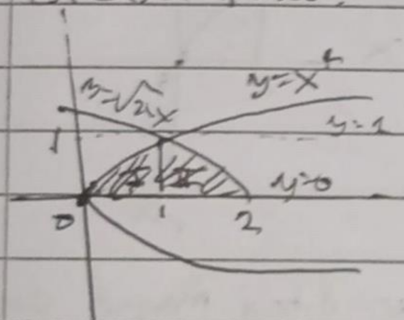
$= \frac{4}{3}\sqrt{y^3} - \frac{1}{2}y^2 \Big|_0^1$

$= \boxed{\frac{5}{6}}$

3 $x = y^4$ dan $y = \sqrt{2-x}$

$y = \sqrt[4]{x}$ dan $y = \sqrt{2-x}$

Sketsa kasar:



Cara I

pecah pakai integral biasa

$\Rightarrow L_I = \int_0^1 \sqrt[4]{x} \, dx$

$L_{II} = \int_1^2 \sqrt{2-x} \, dx$

$\Rightarrow L_T = \frac{4}{5}\sqrt[4]{x^5} \Big|_0^1 + -\frac{2}{3}\sqrt{(2-x)^3} \Big|_1^2$

$= \frac{4}{5} + \left(0 + \frac{2}{3}\right)$

$= \boxed{\frac{22}{15}}$

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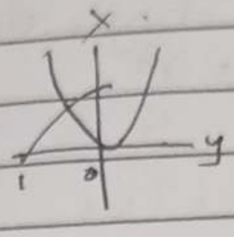
Cara II

Tinjau berdasarkan sumbu y

$x = y^4$ dan $x = 2 - y^2$

dari $y = 0$ sampai $y = 1$



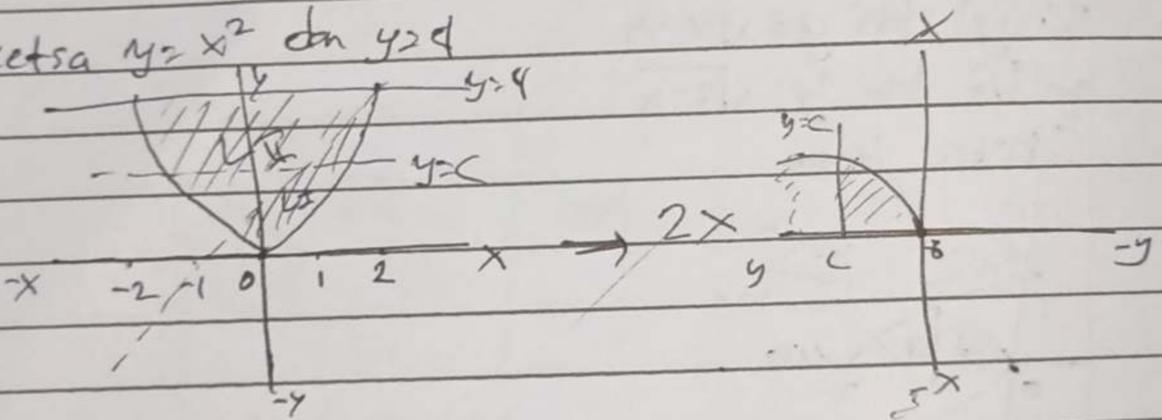
$\int_0^1 ((2-y^2) - y^4)\, dy$

$= 2y - \frac{1}{3}y^3 - \frac{1}{5}y^5 \Big|_0^1$

$= 2 - \frac{1}{3} - \frac{1}{5}$

$= \boxed{\frac{22}{15}}$

4 Sketsa $y = x^2$ dan $y = 4$

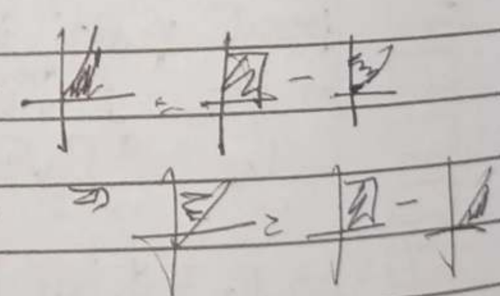


$L_I = L_{II}$, garis dengan $y = c$ agar dibagi menjadi dua

maka

$L_{cx^2} = \frac{1}{2} L_{(-2,2)}$

$\Rightarrow 2\int_0^c \sqrt{y}\, dy = \frac{1}{2}\int_{-2}^{2} x^2\, dx$



$\Rightarrow 2 \cdot \frac{2\sqrt{c^3}}{3} = \left(\frac{1}{2} \cdot 2 \cdot \frac{1}{3} x^3 \Big|_0^2\right) + 8$

$\Rightarrow \frac{4}{3}\sqrt{c^3} = \frac{8}{3} + 8$

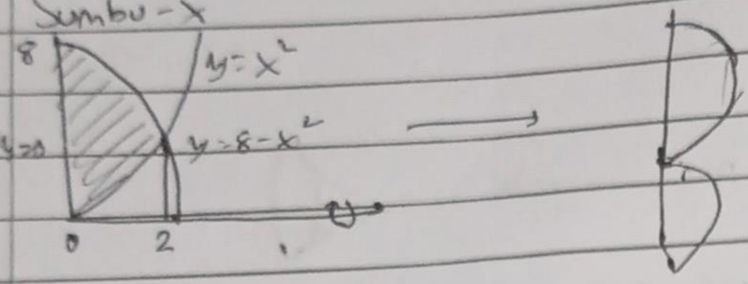
$\Rightarrow c^3 = 16$

$\Rightarrow \boxed{c = \sqrt[3]{16}}$

5 $R: CQ\ I,\ y = 8 - x^2,\ y = x^2,\ y = 0$

Volume pejal mengitari

(a) Sumbu-x



$y_0 = y_1$

$\Rightarrow x^2 = 8 - x^2 \Rightarrow x = 2$ (ambil positif : CQ I)

Maka, volumenya

$$V = \pi \int_0^2 (8-x^2)^2 - (x^2)^2 \, dx$$

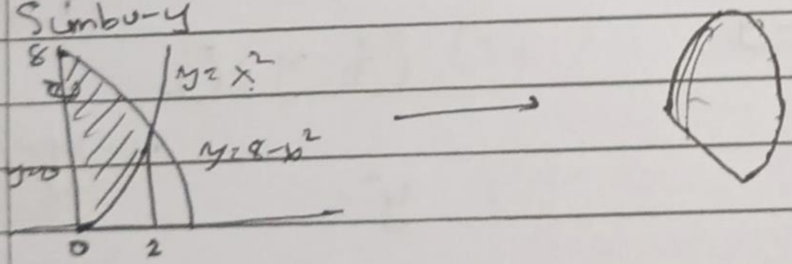
$$= \pi \int_0^2 64 - 32x^2 \, dx$$

$$= 64x - \frac{32}{3}x^3 \Big|_0^2 \cdot \pi$$

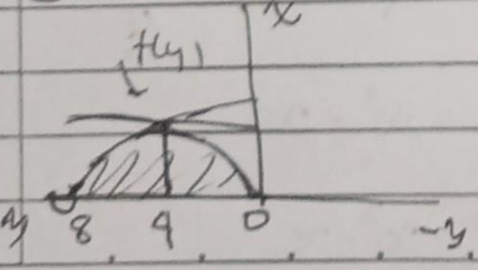
$$= \left(128 - \frac{32}{3} \cdot 2^3\right) \cdot \pi$$

$$= \boxed{\frac{256\pi}{3}}$$

(b) Sumbu-y



Case I



Simetris di/pada $y \in [8, 4]$
(pembuktian dapat mengecek bahwa $f(y-4)$ itu genap)

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Dengan demikian, akan digunakan interval pada $[0, 4]$ dengan $f(y) = \sqrt{y}$, kemudian dikali dua

$$V = \pi \int_0^4 y \, dy \cdot 2$$

$$= 2\pi \frac{1}{2} \cdot y^2 \Big|_0^4$$

$$= \boxed{16\pi}$$

Cara II

Pakai bawang/kulit tabung

$$f(x) = (8 - x^2) - x^2$$

$$= 8 - 2x^2$$

Maka,

$$V = \int_0^2 2\pi x (8 - 2x^2) \, dx$$

$$= 2\pi \int_0^2 8x - 2x^3 \, dx$$

$$= 2\pi \left(4x^2 - \frac{1}{2} x^4\right) \Big|_0^2$$

$$= 2\pi (16 - 8)$$

$$= \boxed{16\pi}$$

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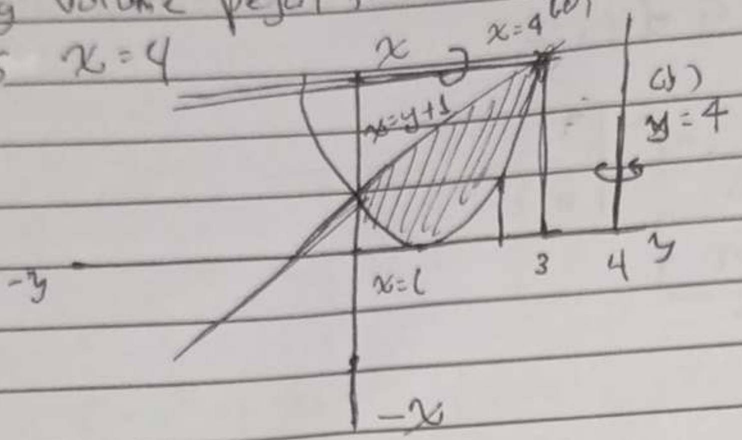
6 R dibatasi kurva-kurva $x=(y-1)^2$ dan $x-y=1$ ↳ $x=y+1$

Hitung volume pejal ketika memutari

(a) garis $x=4$



$x=y+1$ "bawah"

$x=(y-1)^2$ "atas"

↳ karena lebih jauh daripada $x=y+1$ terhadap $x=4$

Menggunakan metode cincin /washer

$$V=\pi\int_0^3 (4-(y-1)^2)-(4-(y+1))\,dy$$

$$=\pi\int_0^3 ((-y^2+2y+3)^2-(3-y)^2)\,dy$$

$$=\pi\int_0^3 y^4-4y^3-3y^2+18y\,dy$$

$$=\pi\left(\frac{1}{5}y^5-y^4-y^3+9y^2\right)\Big|_0^3$$

$$=\boxed{\frac{108\pi}{5}}$$

2

(b) garis $y=4$

Menggunakan metode kulit tabung

$V=2\pi\cdot r\cdot h$

$$=\int_0^3 2\pi(4-y)\cdot((y+1)-(y-1)^2)\,dy$$

$$=\int_0^3 2\pi(y^3-7y^2+12y)\,dy$$

$$=2\pi\left(\frac{1}{4}y^4-\frac{7}{3}y^3+6y^2\right)\Big|_0^3$$

$$=\boxed{\frac{45\pi}{2}}$$

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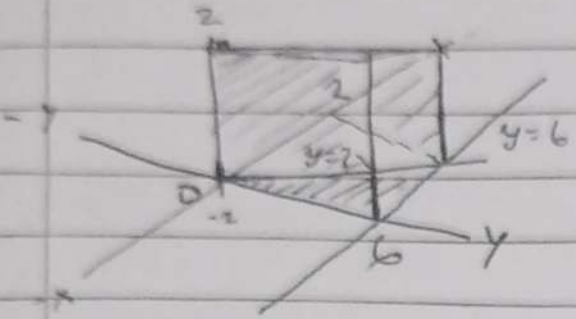

$y_1 = y_2$

$\Rightarrow 6 = 3x$

$\Rightarrow x = 2$

Tidak perlu memakai integral
(Meskipun jawabannya sama)
Perhatikan bahwa alasnya berbentuk segitiga sehingga

$A = \frac{x_{max} y_{max}}{2} = \frac{6 \cdot 2}{2} = 6$

$V = At$
$= 6 \cdot 10$
$= \boxed{60}$

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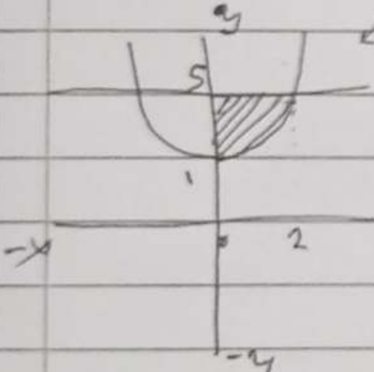
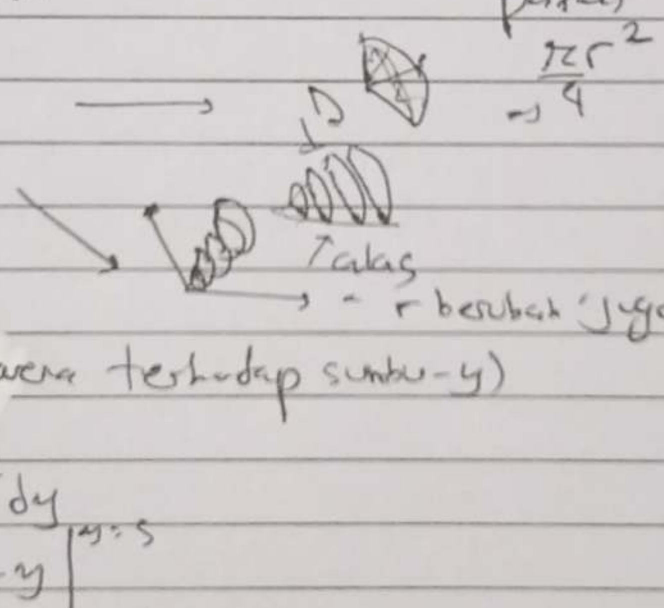


← Sketsa kasar sekali



$r = x = \sqrt{y-1}$ (karena terhadap sumbu-y)

$\Rightarrow V = \int_1^5 \frac{\pi}{4} \cdot r^2 \, dr$

$= \frac{1}{4}\pi \int_1^5 (\sqrt{y-1})^2 \, dy$

$= \frac{1}{4}\pi \left(\frac{1}{2}y^2 - y\right) \Big|_{y=1}^{y=5}$

$= \boxed{2\pi}$

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